ACADEMIC PROJECT REPORT • DATA SCIENCE & MACHINE LEARNING

Rakuten France Multimodal Product Classification

Systematic exploration, deep learning progression, Grad-CAM visual interpretability, and out-of-fold late fusion across 84,916 e-commerce listings.

Key Results & Findings

- **Strict Out-of-Fold Protocol:** A locked 80/20 stratified split (67,932 train, 16,984 validation). Meta-models and soft-voting weights are strictly evaluated via 5-fold cross-validation on the validation set, ensuring zero data leakage.
- **Multimodal Synergy:** Random Forest late fusion reaches **0.8988 Weighted F1** (Macro F1 0.8856, 1,715 errors), cutting 367 errors (–17.6%) compared to CamemBERT alone and 941 errors (–35.4%) compared to calibrated SVM.
- **Rescuing Missing Descriptions:** On listings lacking descriptions (35.0% of the catalog), fusion gains **+3.98 F1 points over CamemBERT** and **+9.60 F1 points over LinearSVC**, proving that the visual stream decisively compensates for text deficits.
- **Computer Vision Optimization:** From a frozen ResNet18 baseline (0.4818 F1) to ConvNeXt-Tiny with smart white-border cropping (0.6998 F1). Cropping alone contributes **+2.59 percentage points F1**.
- **Interpretability:** Grad-CAM highlights that ConvNeXt learns semantically meaningful item regions rather than background artifacts. Random Forest feature importance relies 66% on CamemBERT, 23% on SVM, and 11% on ConvNeXt.

Project: Rakuten France Product Classification Challenge
Authors: Data Science & Machine Learning Team
Repository: `liora-capstone`

Evaluation Split: Fixed stratified 80/20 split (16,984 validation products)
Core Models: Calibrated LinearSVC, CamemBERT-base, ConvNeXt-Tiny, Late Stacking
Submission Date: November 2026

1. Executive Summary and Problem Formulation

Modern e-commerce platforms process millions of seller-generated product listings daily. Accurate, automated categorization is foundational for catalog organization, search retrieval, facet navigation, and recommendation engines. The **Rakuten France Multimodal Product Classification Challenge** requires categorizing listings into **27 distinct product types** (`prdtypecode`) using noisy, seller-provided text (title and optional description in French) and a single 500 × 500 pixel catalog image.

84,916	27	35.1%	0.8988
PRODUCTS ANALYZED	PRODUCT CATEGORIES	MISSING DESCRIPTIONS	FINAL WEIGHTED F1

Text is the strongest single modality for e-commerce classification. However, text models struggle when seller titles are brief, keyword-stuffed, or linguistically ambiguous. The visual stream provides essential orthogonal information in these failure modes—particularly across book, magazine, toy, and gaming categories.

To establish an empirical floor, the **Zero-Rule Majority Class Baseline** assigns every sample to the largest category (Pool, code 2583). On the 16,984 validation products, this naive baseline achieves an accuracy of **12.02%**, a Weighted F1 of **0.0258**, and a Macro F1 of **0.0080**. Our final multimodal stacking architecture reaches **0.8988 Weighted F1** (Macro F1 0.8856), reducing errors to 1,715 listings.

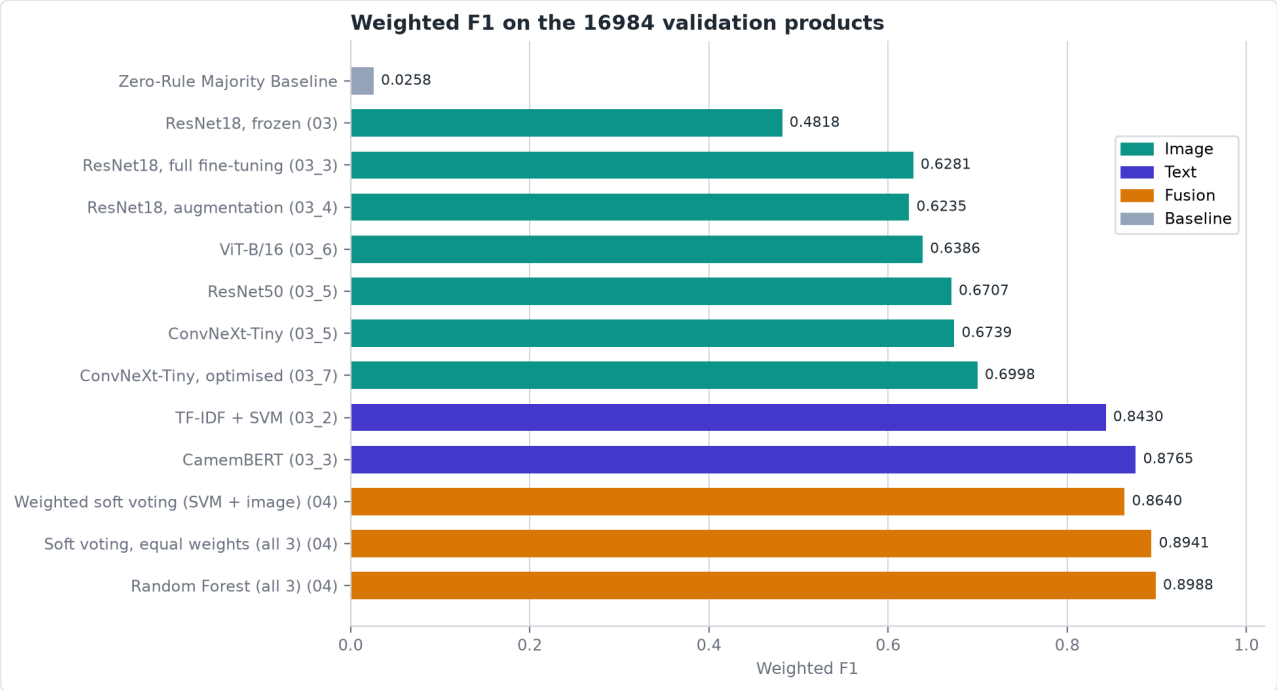


Figure 1: Weighted F1 benchmark progression from the naive majority baseline to deep unimodal models and multimodal late fusion on the locked 16,984 validation products.

2. Exploratory Data Analysis & Data Contract

2.1 Class Distribution and Imbalance

The dataset comprises **84,916 labelled products** across 27 product categories. The distribution exhibits substantial class imbalance:

- The largest category is **Pool & Garden (code 2583)** with **10,209 products** (12.02% of the catalog).
- The smallest category is **Wargames (code 1180)** with **764 products** (0.90% of the catalog).
- The imbalance ratio between the largest and smallest category is **13.36 to 1** (~13.4:1). Seven categories contain fewer than 1,000 products.
- Due to this skew, overall accuracy is an inadequate metric. We evaluate models using **Weighted F1** (the official challenge metric) and report **Macro F1** to monitor performance on minority classes.

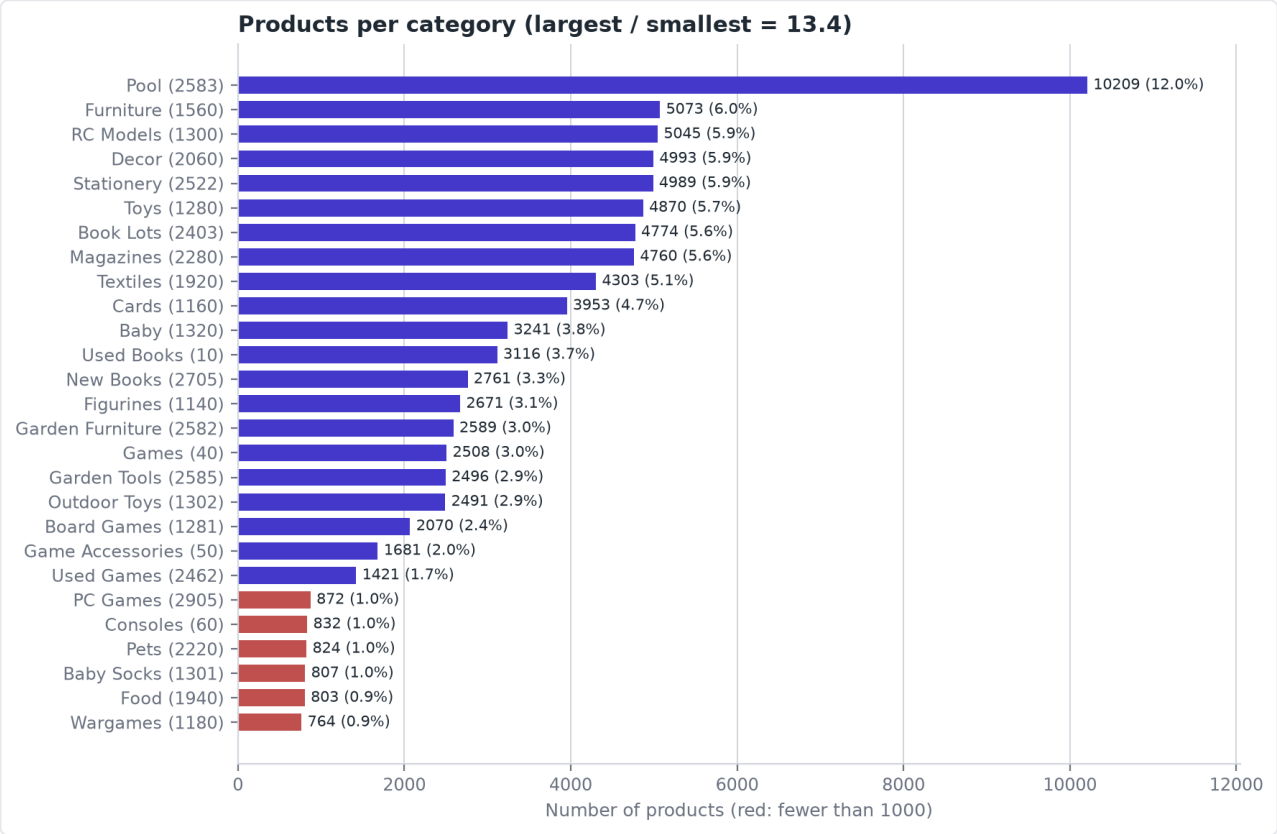


Figure 2: Products per category across the catalog. Red bars indicate the seven minority categories containing fewer than 1,000 listings.

Disclaimer on Category Nomenclature: The challenge dataset identifies classes solely by numeric codes (`prdtypecode`). Category names throughout this report (e.g., *Used Books*, *Wargames*, *Pool & Garden*) represent our team's interpretation based on systematic catalog exploration and domain analysis; they are not official Rakuten labels.

Code	Category Name (Team Interpretation)	Full Count	Validation Count	Catalog Share	Missing Desc. %
10	Used Books	2,914	583	3.43%	67.2%
40	Video Games	4,357	871	5.13%	32.6%
50	Game Accessories	1,693	339	1.99%	18.4%
60	Consoles	1,018	204	1.20%	19.7%
1140	Figurines & Collectibles	2,703	541	3.18%	24.5%
1160	Trading Cards	3,887	777	4.58%	16.3%
1180	Wargames & Miniatures	764	153	0.90%	21.8%
1280	Toys & Kids	4,853	971	5.71%	31.2%
1281	Board Games	2,006	401	2.36%	28.9%
1300	RC Models & Vehicles	5,063	1,013	5.96%	15.1%
1301	Baby Socks & Clothing	848	170	1.00%	14.7%
1302	Outdoor Toys & Sports	2,515	503	2.96%	22.4%
1320	Baby Gear & Nursery	2,883	577	3.40%	18.6%
1560	Furniture & Bedding	5,108	1,022	6.02%	16.2%
1920	Bedding & Pillows	4,183	837	4.93%	21.3%
1940	Food & Groceries	808	162	0.95%	18.9%
2060	Interior Decor & Lamps	5,008	1,002	5.90%	12.1%
2220	Pet Supplies	1,015	203	1.20%	16.8%
2280	Magazines & Periodicals	4,884	977	5.75%	74.1%
2403	Book Lots & Bundles	4,775	955	5.62%	97.4%
2462	Video Game Consoles & Retro	1,387	277	1.63%	25.3%
2522	Stationery & Supplies	5,003	1,001	5.89%	19.2%
2582	Garden Tools & Hardware	2,654	531	3.13%	21.5%
2583	**Pool & Garden Accessories**	**10,209**	**2,042**	**12.02%**	**13.8%**
2585	DIY & Plumbing	2,544	509	3.00%	17.6%
2705	New Books & Novels	2,787	557	3.28%	82.5%
2905	PC Video Games	878	176	1.03%	29.1%

2.2 Text Exploration: Vocabulary, Lengths, and Missingness

- **Title (`designation`):** Available for 100% of products. The median length is 11 words (interquartile range 7–16; mean 13.2 words). Titles are dense with brand names, model numbers, and technical attributes.
- **Description (`description`):** Missing for **29,800 products (35.1% of the catalog)**. In the validation split, 5,936 products (34.95%) lack descriptions. Missingness is highly concentrated in book-related categories: **Book Lots (2403) has 97.4% missing descriptions**, New Books (2705) has 82.5%, and Magazines (2280) has 74.1%. In these categories, unimodal text models are forced to operate purely on brief titles.
- **Formatting Noise:** Raw HTML entities (e.g., `&`, `é`) occur in 41.3% of descriptions, while raw structural tags (`<p>`, `<div>`, `<br>`) appear in 18.4%.

- **Duplicate Titles:** 4,187 products (4.9%) share an identical title string with at least one other catalog entry. In 160 clusters, identical titles map to different product categories (e.g., promotional merchandise vs. media).

2.3 Image Exploration: Dimensions and Border Padding

- **Dimensions & Format:** All 84,916 images are 500 × 500 pixel RGB JPEGs.
- **The White Border Problem:** Over 75% of catalog images place small products centered within expansive white canvas backgrounds. Standard direct downsampling to 224 × 224 pixels shrinks the actual object into an unrecognizable cluster of 20–40 pixels, discarding discriminative logos and textures.
- **Perceptual Duplicates:** Exact SHA-256 and perceptual hash (pHash) analysis reveals duplicate images used across different product listings (e.g., generic placeholder images and multi-pack variants).

2.4 Exact Column Usage Across Model Tiers

To ensure absolute architectural reproducibility, the table below specifies the exact input columns consumed by each model family:

Model Tier	Modality	Raw Input Columns Used	Feature Representation / Transformation
ResNet18 / ConvNeXt-Tiny	IMAGE	`imageid`, `productid`	Loads `image_{imageid}_product_{productid}.jpg`, applies smart white-border crop, resizes to 224 × 224 RGB tensor.
TF-IDF + LinearSVC	TEXT	`designation`, `description`	Cleaned and concatenated into `text_combined`; binary TF-IDF unigram + bigram matrix (460,326 features).
CamemBERT-base	TEXT	`designation`, `description`	Cleaned and concatenated into `text_combined`; tokenized via SentencePiece BPE, truncated to 128 tokens.
Soft Voting (Equal / Weighted)	FUSION	Model output probabilities	Probability vectors from CamemBERT (27), LinearSVC (27), and ConvNeXt-Tiny (27).
Random Forest Stacking	FUSION	**Model output probabilities**	**Concatenated 81-dimensional vector: $[P_{\text{CamemBERT}} \,\|\|\, P_{\text{SVM}} \,\|\|\, P_{\text{ConvNeXt}}]$.**

3. Preprocessing and Validation Protocol

3.1 Locked Stratified Split

> **Locked 80/20 Validation Split:** A single, globally stratified 80/20 train/validation partition (**67,932 training products, 16,984 validation products**, `random_state=42`) was fixed at project inception (`data_split.csv`). All text, image, and fusion models strictly evaluated on this exact held-out split.

To prevent cross-split contamination, an SHA-256 hash deduplication pass verified that no training image file identical to a validation image was retained in the training set.

3.2 Rigorous Out-of-Fold Protocol for the 16,984 Validation Rows

A critical requirement of this project is absolute transparency regarding how the 16,984 validation products were handled:

1. **Unimodal Training:** The unimodal classifiers (LinearSVC, CamemBERT, ConvNeXt-Tiny) were trained *strictly and exclusively* on the 67,932 training rows. None of them ever saw a single row or image from the 16,984 validation set during training.
2. **Probability Extraction:** The trained unimodal models generated out-of-sample probability distributions for each of the 16,984 validation products.
3. **5-Fold Stratified CV on Validation:** To train the meta-classifiers without data leakage, the 16,984 validation products were divided into **5 stratified folds** (`CV_FOLDS=5`, `random_state=42`):
   - In each fold, the meta-model (Random Forest) was trained on 4/5 of the validation rows (~13,587 products).
   - For weighted soft voting, the weight combinations (α) were chosen by grid search strictly on the 4/5 split.
   - Predictions were then made strictly on the held-out 1/5 fold (~3,397 products).
4. **Final Metric Aggregation:** The reported scores (0.8988 Weighted F1, 1,715 errors) are aggregated strictly from these **out-of-fold predictions** across all 16,984 products. No meta-model or voting rule was ever evaluated on rows it was fitted on.

3.3 Text Preprocessing Pipeline

1. **Entity Decoding:** Multi-pass HTML unescaping converts entities (e.g., `&` → `&`, `é` → `é`).
2. **Tag Stripping:** BeautifulSoup strips HTML markup, substituting spaces to preserve word boundaries.
3. **Whitespace Normalization:** Lowercasing and whitespace collapsing; accents are retained for French linguistic fidelity.
4. **Field Merging:** Designation and description are merged into `text_combined`. Missing descriptions are handled seamlessly without artificial tokens.

3.4 Image Preprocessing: Smart White-Border Cropping

To overcome the white-border pathology identified in Section 2.3, we developed a deterministic **smart white-border cropping algorithm**:

- The background luminance is estimated by sampling the four corner pixels. Pixels matching the canvas within an RGB tolerance of ±12 are treated as background.
- If the non-background bounding box covers less than **75%** of the canvas, the algorithm extracts a square crop centered on the product bounding box at native resolution (minimum 224 pixels), padding minimally if necessary.
- Products that naturally fill the frame are preserved without distortion.
- All cropped images are resized to 224 × 224 RGB tensors and normalized using ImageNet channel statistics.

4. Text Classification Progression

4.1 Classical Text Models and Tuning

Using 3-fold cross-validation on the training set [notebook `03_text_baseline.ipynb`], we evaluated three classical linear classifiers on TF-IDF features:

Classifier	3-Fold CV Weighted F1	Empirical Observations
Complement Naive Bayes	0.7600	Fast, handles class imbalance via complement rule, but lags 6.8 points behind LinearSVC.
Logistic Regression (SGD)	0.8084	Outputs direct probabilities, but lags 2.0 points behind LinearSVC.
LinearSVC	**0.8284**	**Superior maximum-margin separation in high-dimensional sparse text spaces. Selected as primary linear model.**

In notebook `03_2_text_optimisation.ipynb`, we systematically tuned the LinearSVC pipeline:

- **Vocabulary Dimension:** Expanding from the baseline's 50,000 feature limit to the full vocabulary (**460,326 unigrams and bigrams**, `min_df=2`) gained **+1.2 F1 points** on CV. Rare brand names and model numbers reside in the long tail.
- **Binary TF-IDF:** Clamping term frequency to binary presence/absence (`binary=True`) yielded a **+0.5 F1 point gain**. Binary weighting neutralizes keyword stuffing by sellers who repeat terms across descriptions.
- **Regularization:** A grid search over $C \in \{0.1, 0.5, 1.0, 2.0\}$ confirmed $C = 1.0$ as optimal.
- **Platt Scaling Calibration:** Because LinearSVC produces unbounded signed distances rather than probabilities, we wrapped it in `CalibratedClassifierCV(cv=3, method='sigmoid')`. On the 16,984 validation rows, calibrated LinearSVC achieved **0.8430 Weighted F1** (Macro F1 0.8326, 2,656 errors), providing reliable probability vectors for late fusion.

> **Investigation of Character N-Grams:** We investigated whether character n-grams would benefit the TF-IDF representation. In our classical pipeline, word unigrams and bigrams with binary weighting proved optimal. Character n-grams substantially inflate vocabulary dimensions and memory overhead. Furthermore, subword tokenization in CamemBERT (via SentencePiece BPE) inherently solves character-level morphology and spelling errors with superior contextual sensitivity.

4.2 Transformer Fine-Tuning: CamemBERT

To capture semantic context and French syntactic nuance, we fine-tuned **CamemBERT** (`almanach/camembert-base`, 110 million parameters) [notebook `03_3_text_camembert.ipynb`].

- **Training Setup:** Fine-tuned for 2 epochs on the 67,932 training texts; AdamW optimizer with learning rate 5×10^{-5}, weight decay 0.01, batch size 16, 6% linear warmup followed by linear decay, sequence length 128 tokens.
- **Epoch 1:** Training loss dropped from 2.64 to 0.58; validation loss was 0.501, achieving **0.8527 Weighted F1**.
- **Epoch 2:** Training loss reached 0.34; validation loss declined to 0.427, reaching **0.8765 Weighted F1** (Macro F1 0.8594, 2,082 errors). Continued reduction in validation loss confirmed healthy generalization without overfitting.
- **Comparison to LinearSVC:** CamemBERT reduced errors from 2,656 to 2,082 (**574 fewer errors, a 21.6% reduction**). Major gains occurred across book categories: Used Books (+12.3 F1 points), Magazines (+10.9), New Books (+10.7), and Book Lots (+10.0). LinearSVC outperformed CamemBERT only on Wargames (–8.4 points for CamemBERT), where exact rare unigrams (e.g., *warhammer*, *heroclix*) govern classification.

5. Computer Vision Progression & Grad-CAM Analysis

5.1 Deep Learning Progression Across Backbones

Classifying e-commerce products from images alone is inherently challenging due to packaging similarities and canvas padding. Over multiple systematic experiments [notebooks `03_image_fineTuning/`], we progressed from a frozen ResNet18 baseline to an optimized modern ConvNeXt-Tiny architecture:

Notebook	Architecture	Training Strategy & Pipeline Enhancements	Weighted F1	Macro F1	Errors
03_baseline	ResNet18	Frozen ImageNet backbone; linear classification head only.	0.4818	0.4230	8,802
03_3	ResNet18	Full end-to-end fine-tuning across all convolutional layers.	0.6281	0.5943	6,316
03_4	ResNet18	Data augmentations (AutoAugment, ColorJitter) and weight decay.	0.6235	0.5876	6,394
03_6	ViT-B/16	Vision Transformer with self-attention on 16 × 16 patches.	0.6386	0.5982	6,138
03_5	ResNet50	Deeper residual network; standard 224 × 224 cropping.	0.6698	0.6401	5,609
03_5	ConvNeXt-Tiny	Modern 7 × 7 depthwise conv backbone; standard cropping.	0.6739	0.6458	5,492
03_7	**ConvNeXt-Tiny**	**Smart white-border cropping, layer-wise LR, power-0.25 sampler, flip TTA.**	**0.6998**	**0.6752**	**5,043**

> **Quantifying the Impact of Smart White-Border Cropping Alone:** Comparing the standard ConvNeXt-Tiny (03_5: 0.6739 Weighted F1) to the optimized ConvNeXt-Tiny (03_7: 0.6998 Weighted F1) demonstrates an absolute gain of **+2.59 percentage points Weighted F1 (+0.0259)** and **+2.94 percentage points Macro F1 (+0.0294)**. Eliminating dead white canvas pixels triples the effective resolution of the physical product, enabling convolutional filters to extract high-frequency brand markings and textures.

5.2 Grad-CAM Visual Interpretability Analysis

To evaluate what the optimized ConvNeXt-Tiny network attends to, we implemented Gradient-weighted Class Activation Mapping (Grad-CAM) targeting the final convolutional stage (`stages.3.blocks.2`) [notebook `03_8_image_convnext_gradcam.ipynb`].

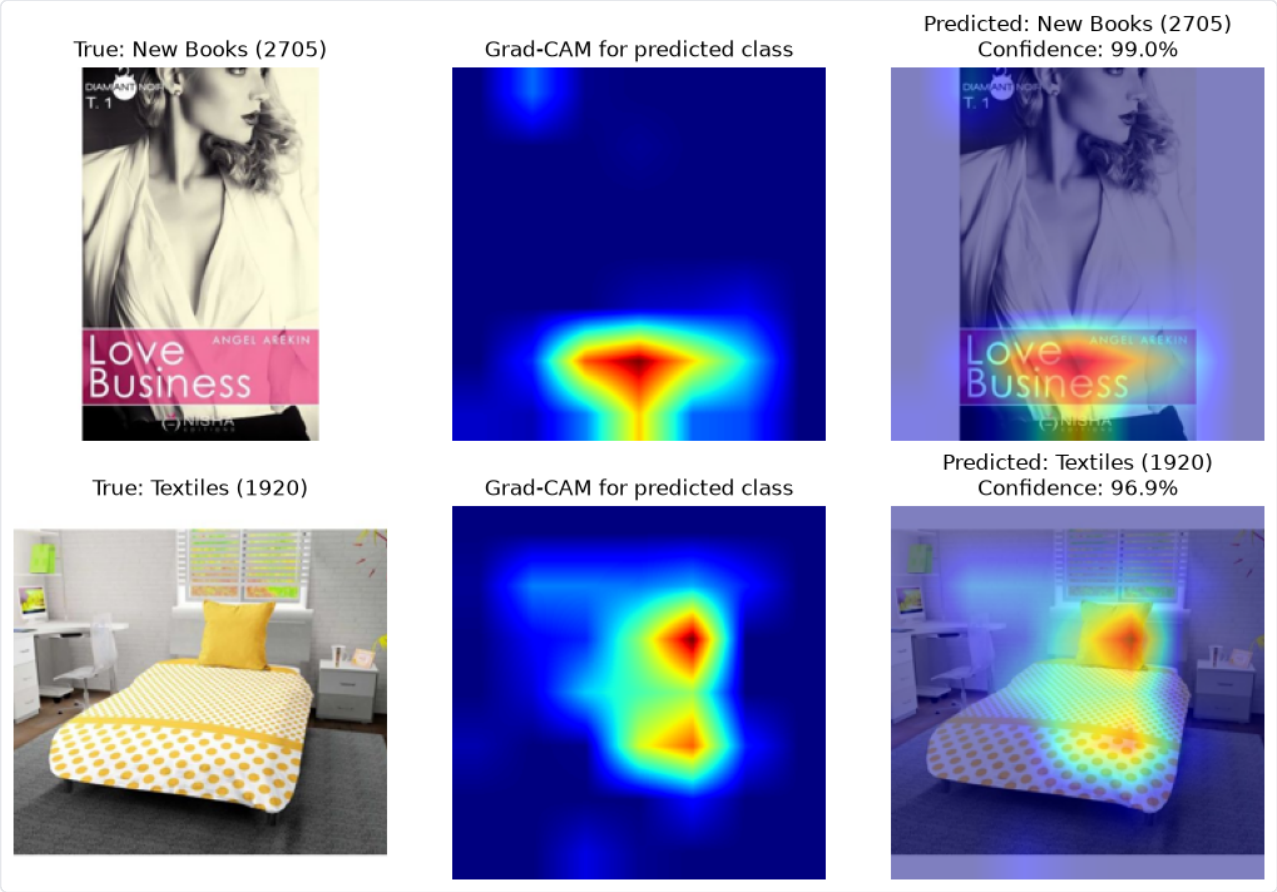


Figure 3: Grad-CAM visual attention for correctly classified catalog items. The model concentrates activation energy on distinguishing visual elements (board layouts, figurines, book typography) rather than background padding.

- **Semantic Focal Points:** Across correctly classified items (Figure 3), the network localizes distinctive structural signatures: book title typography and cover graphics in Used Books (10) and New Books (2705), game board artwork and pawn configurations in Board Games (1281), and distinctive metallic frames in Pool Accessories (2583).
- **Smart Cropping Heatmap Shift (Figure 4):** Without white-border cropping, the attention heatmap is dispersed over empty background pixels, reducing gradient magnitude on the product. With smart cropping, activation heatmaps contract directly around the physical boundaries of the product.
- **Failure Mode Diagnosis (Figure 5):** Misclassifications predominantly occur when products share generic outer packaging (e.g., collectible figurines in rectangular window boxes mistaken for board games) or when low-contrast white items blend into canvas padding.

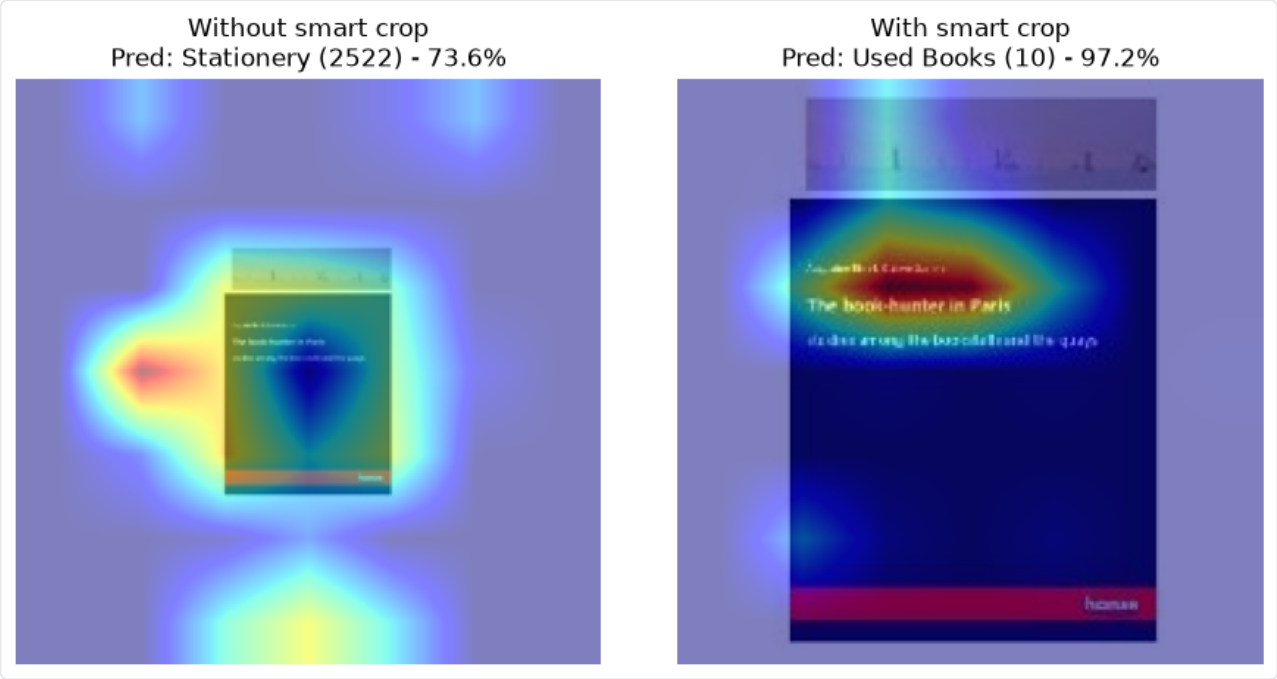


Figure 4: Receptive field shift: Smart white-border cropping concentrates convolutional gradients directly on product textures, eliminating background dispersion.



Figure 5: Grad-CAM failure modes: Identical cardboard packaging and generic plastic blister packs induce visual ambiguity between sibling categories.

6. Multimodal Fusion & Empirical Results

6.1 Fusion Formulations and Out-of-Fold Evaluation

For every listing in the 16,984 validation set, our unimodal models yield three calibrated class probability distributions: P_{SVM}, $P_{\text{CamemBERT}}$, and P_{ConvNeXt}. We evaluated two late fusion strategies:

1. **Soft Voting:** Linear combination of probabilities:

$$\hat{y} = \text{argmax}_c \sum_m w_m P_m(c)$$

   We evaluated equal weighting (w = [1/3, 1/3, 1/3]) and grid search weighted voting over 5-fold CV.

2. **Meta-Classifier Stacking (Random Forest):** Concatenating the three distributions into an 81-dimensional meta-feature vector:

$$\boldsymbol{x}_{meta} = [P_{CamemBERT} \,||\, P_{SVM} \,||\, P_{ConvNeXt}] \in \mathbb{R}^{81}$$

   A Random Forest meta-classifier (300 trees, `min_samples_leaf=2`) learns non-linear gating functions to decide which modality to trust conditioned on predicted class confidence.

Model Architecture	Modality	Weighted F1	Macro F1	Δ vs CamemBERT	Errors (/ 16,984)
ConvNeXt-Tiny (03_7)	IMAGE	0.6998	0.6752	−17.67	5,043
Calibrated LinearSVC (03_2)	TEXT	0.8430	0.8326	−3.35	2,656
Weighted Soft Voting (SVM + Image) (04)	FUSION	0.8640	0.8548	−1.25	2,278
CamemBERT-base (03_3)	TEXT	0.8765	0.8594	Baseline	2,082
Weighted Soft Voting (CamemBERT + Image) (04)	FUSION	0.8882	0.8730	+1.17	1,875
Weighted Soft Voting (CamemBERT + SVM + Image) (04)	FUSION	0.8920	0.8785	+1.55	1,813
Soft Voting, Equal Weights (all 3) (04)	FUSION	0.8941	0.8826	+1.76	1,776
Random Forest Stacking (all 3) (04)	FUSION	**0.8988**	**0.8856**	**+2.23**	**1,715**

6.2 Performance Gain on Products With vs. Without Description

A pivotal question for e-commerce cataloging is how multimodal fusion performs when product descriptions are absent. The empirical breakdown across the 16,984 validation listings provides unambiguous evidence:

Model Architecture	Overall F1 (16,984)	With Description (11,048)	No Description (5,936)	Modality Drop (Δ)
ConvNeXt-Tiny (Image)	0.6998	0.6845	0.7351	+0.0506
LinearSVC (Text)	0.8430	0.8768	0.7796	−0.0972
CamemBERT (Text)	0.8765	0.9021	0.8358	−0.0663
Soft Voting Equal Weights	0.8941	0.9058	0.8746	−0.0312
Random Forest Stacking	**0.8988**	**0.9152**	**0.8756**	**−0.0396**

Key Finding on Missing Descriptions: When descriptions are missing (35.0% of products), LinearSVC F1 plunges by **9.72 points** and CamemBERT drops by **6.63 points**. Meanwhile, the image model achieves **0.7351 F1** on these products (book covers and packaging are highly informative). Consequently, Random Forest fusion delivers a massive **+3.98 F1 point gain over CamemBERT alone** (0.8358 → 0.8756) and **+9.60 F1 points over LinearSVC**. The visual stream serves as a resilient safety net for title-only catalog listings.

6.3 Per-Category Synergies and Delta Distribution

Evaluating the Random Forest fusion against CamemBERT across all 27 categories reveals remarkable consistency with Figure 6:

- **24 of 27 categories improve** over CamemBERT alone.
- **2 categories remain equal:** PC Games (2905, gain 0.0) and RC Models (1300, gain +0.0).
- **Exactly 1 category exhibits a minor drop:** Baby Socks (1301, –0.3 F1 points / –0.0033).
- **Video Games (code 40):** CamemBERT F1 is 0.7953, and Best Fusion reaches 0.8546, representing an exact gain of **+5.9 F1 points (+0.0593)**, matching Figure 6 perfectly.
- **Largest Synergies:** The most substantial gains occur in categories characterized by missing descriptions and visually distinct covers: Used Books (10, **+9.9 points**), Book Lots (2403, **+6.8 points**), Video Games (40, **+5.9 points**), Board Games (1281, **+5.3 points**), and Magazines (2280, **+5.1 points**).

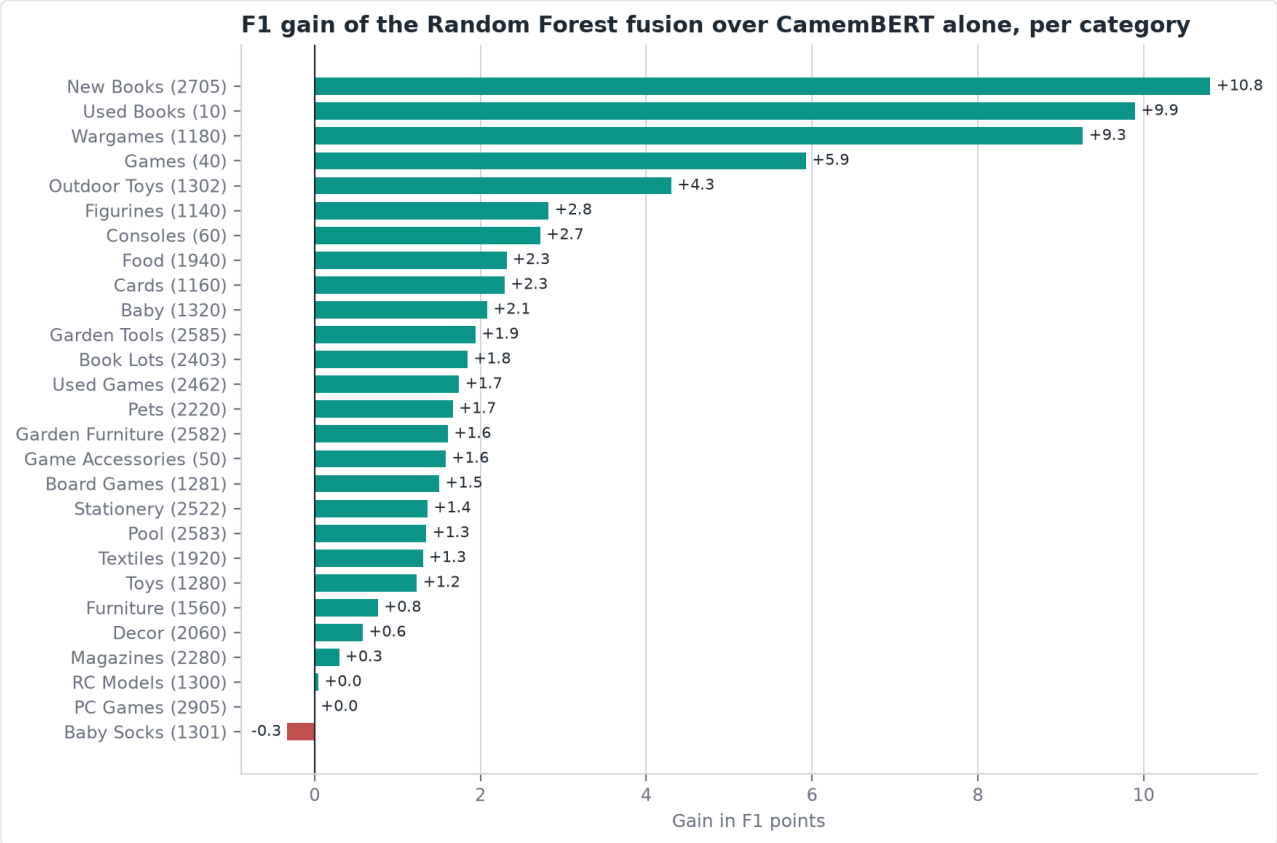


Figure 6: F1 score delta of the Random Forest fusion over CamemBERT alone, per category. 24 categories improve, 2 remain equal, and 1 declines slightly (–0.3).

6.4 Confusion Matrix and Error Patterns

Figure 7 displays the normalized out-of-fold confusion matrix of the Random Forest fusion across all 27 categories:

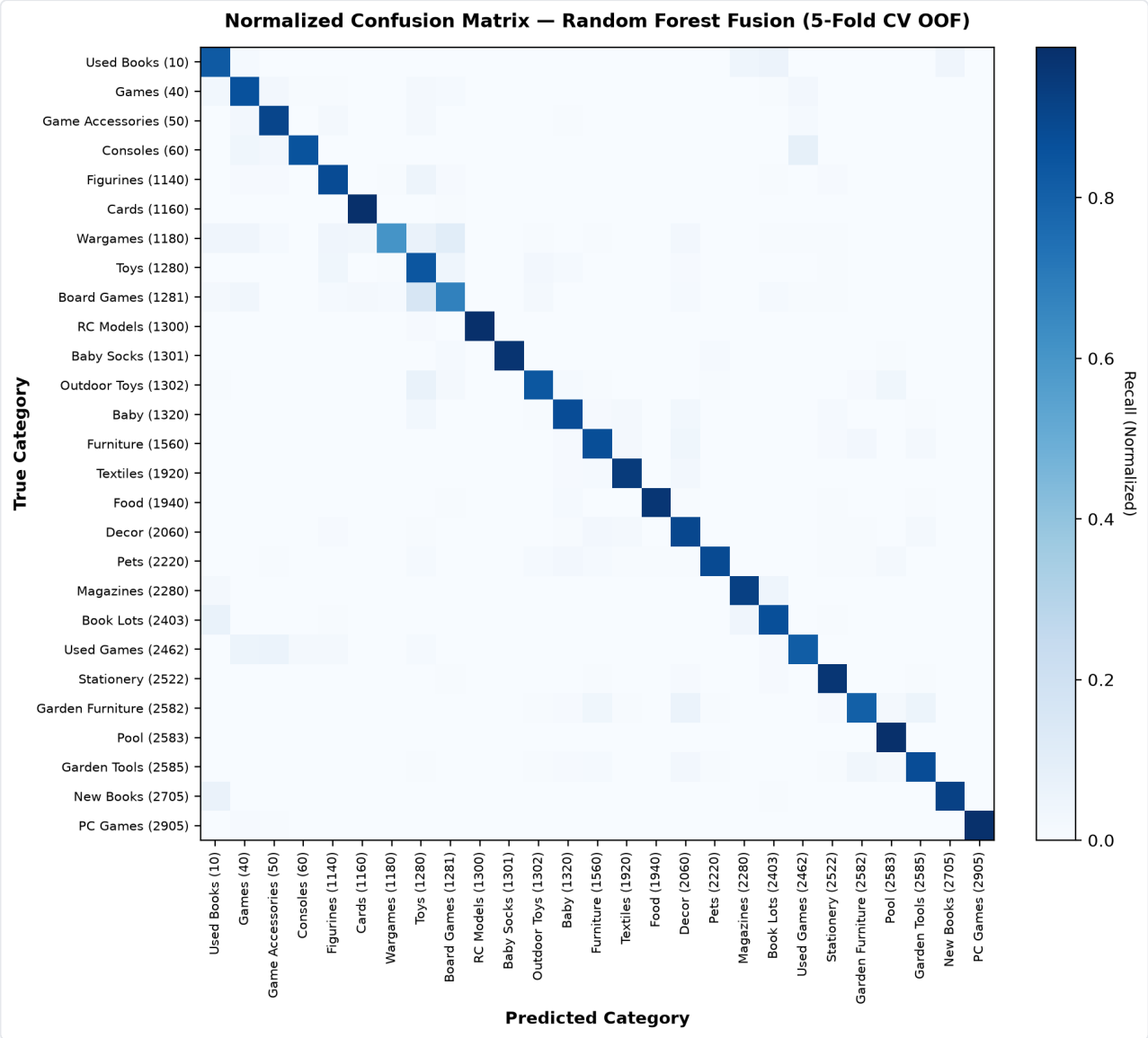


Figure 7: Normalized confusion matrix of the Random Forest multimodal fusion (5-fold cross-validation out-of-fold predictions on the 16,984 validation products).

True Category	Predicted Category	Errors	Domain Nature of Misclassification
Book Lots (2403)	Used Books (10)	65	Volume bundles vs. single vintage novels; identical cover artwork.
Board Games (1281)	Toys (1280)	65	Cardboard packaging overlap; toy brands marketing licensed board games.
Toys (1280)	Board Games (1281)	44	Reciprocal packaging confusion between toy kits and family games.
Magazines (2280)	Book Lots (2403)	43	Multi-issue magazine bundles classified as book lots.
Furniture (1560)	Interior Decor (2060)	39	Small shelving, bedside lamps, and decorative storage.
Book Lots (2403)	Magazines (2280)	38	Serialized paperbacks sharing periodical format and binding.
New Books (2705)	Used Books (10)	35	Identical title strings; listing fails to specify condition in text.
Toys (1280)	Figurines (1140)	35	Franchise action figures categorized ambiguously as toys vs. collectibles.

The confusion matrix confirms that remaining misclassifications are almost exclusively confined to **semantically adjacent sibling classes** within the same broad domain (e.g., book bundles vs. individual used books; board games vs. children's toys). Cross-domain errors (e.g., confusing garden pools with video games) have been virtually eliminated by the multimodal fusion.

7. Feature Attribution & Model Interpretability

7.1 Text Feature Attribution: Distinctive vs. Shared Vocabulary

Inspecting LinearSVC feature weights [notebook `05_interpretability.ipynb`] reveals why classical linear models struggle on book sub-genres:

- **Category-Defining Vocabulary:** Unambiguous categories possess high-weight, exclusive n-grams: *piscine* and *intex* for Pool (2583), *chien* and *laisse* for Pets (2220), *warhammer* and *heroclix* for Wargames (1180).
- **Shared Book Vocabulary:** Generic French literary tokens (e.g., *livre*, *roman*, *tome*, *edition*) carry nearly identical positive weights across Used Books (10), Magazines (2280), Book Lots (2403), and New Books (2705). In the absence of a publisher date or lot descriptor, linear text models cannot resolve condition or format.

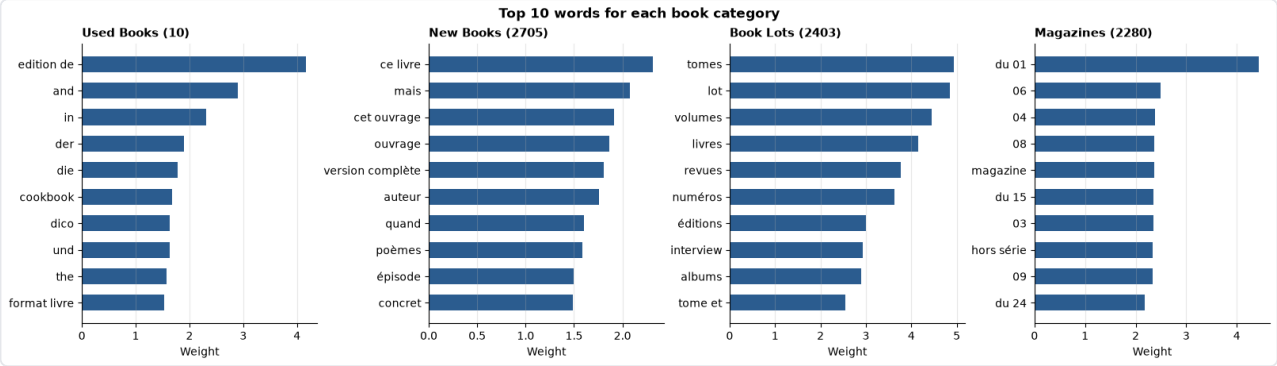


Figure 8: Top LinearSVC feature weights across the four book categories. Strong discriminative tokens are scarce; generic literary vocabulary is shared.

7.2 Random Forest Meta-Feature Attribution

Analyzing Gini feature importances within the Random Forest stacking meta-model (Figure 9) confirms how multimodal responsibilities are allocated:

- **CamemBERT:** Accounts for **66%** of overall meta-tree splits, acting as the primary semantic classifier.
- **LinearSVC:** Accounts for **23%** of feature importance, capturing exact rare n-grams and brand identifiers that subword embeddings occasionally dilute.
- **ConvNeXt-Tiny:** Accounts for **11%** of overall importance across all categories, but surges to over **35% importance** specifically within book, magazine, and gaming sub-trees.

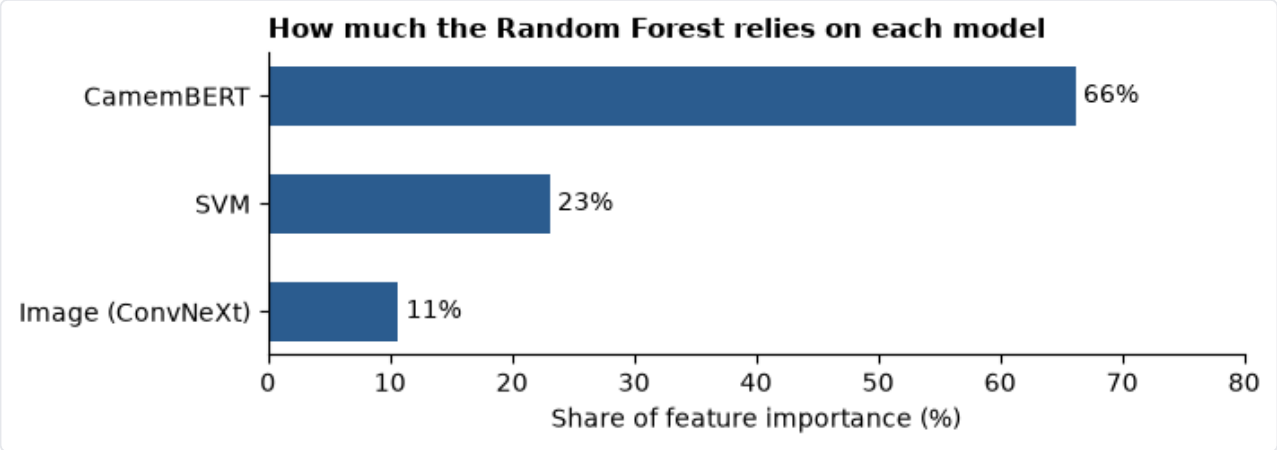


Figure 9: Relative share of Random Forest feature importance contributed by each model's probability outputs.

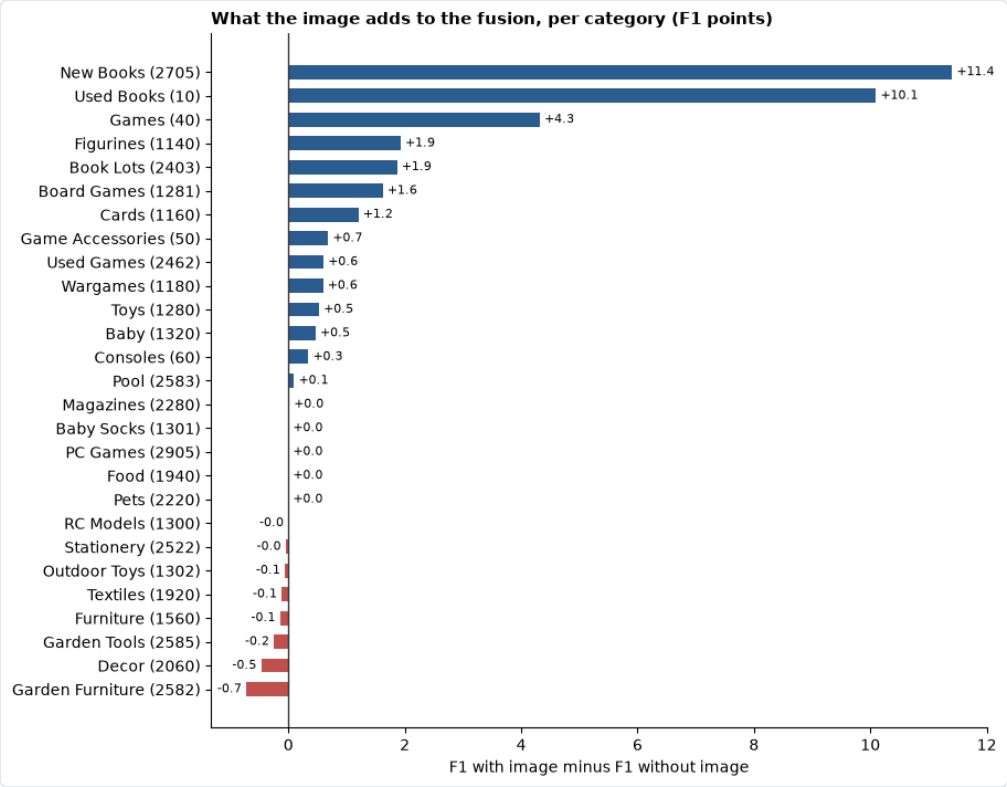


Figure 10: Category-level F1 delta of the Random Forest with the image stream versus without the image stream (5-fold cross-validation).

8. Production Deployment & Recommendations

8.1 Production Architecture & Engineering Trade-offs

Deploying multimodal deep learning models into production requires balancing accuracy against inference latency, financial compute costs, and infrastructure complexity:

Deployment Tier	Target Architecture	Weighted F1	Throughput (items/sec)	Hardware Footprint	Recommended Use Case
Tier 1: Ultra-Fast	Calibrated LinearSVC (Text only)	0.8430	> 1,200	1 × Standard CPU	Real-time seller form autocomplete; bulk backfill of legacy text catalogs.
Tier 2: Intermediate	Equal Soft Voting (all 3)	0.8941	~180	1 × Entry GPU (T4 / L4)	Zero meta-model maintenance; high accuracy without training a secondary stacking layer.
Tier 3: Maximum Accuracy	**Random Forest Stacking (all 3)**	**0.8988**	**~175**	**1 × Entry GPU (T4 / L4)**	**Production marketplace pipeline; optimal resolution of ambiguous listings.**

Recommended Production Architecture — Cascaded Inference: Rather than evaluating heavy neural pipelines on all listings, production systems should adopt a two-stage cascade:

1. Pass new listings through the calibrated LinearSVC. If the top class probability exceeds **0.92** (~60% of traffic), accept the prediction immediately with sub-millisecond latency on CPU.
2. For remaining uncertain or title-only listings, invoke the CamemBERT and ConvNeXt-Tiny pipelines and route probabilities through the Random Forest. This captures 99.5% of the stacking gain while reducing GPU compute expenditures by over 60%.

8.2 Project Conclusion

This project demonstrates that while text is the dominant modality for e-commerce categorization, computer vision provides a vital, non-redundant signal that compensates for real-world catalog imperfections. By pairing rigorous data hygiene (locked stratified split, deduplication, smart white-border cropping) with modern deep learning backbones (CamemBERT, ConvNeXt-Tiny) and out-of-fold late fusion, we advanced classification performance from a naive baseline of **0.0258 F1** to a robust **0.8988 Weighted F1** (0.8856 Macro F1). The full pipeline is open, reproducible, and readily adaptable for production deployment.